

GRAND RAPIDS FORD, MI, USA

WMO: 726350

Lat: 42.883N

Lon: 85.524W

Elev: 245

StdP: 98.42

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94860

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-14.0	-20.5	0.6	-15.5	-18.0	0.8	-13.2	12.2	-2.8	11.2	-3.1	3.7	200	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	31.9	22.7	30.4	22.0	29.0	21.2	24.6	29.6	23.6	28.2	22.7	27.1	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.3	27.3	22.0	17.2	26.1	21.1	16.3	25.3	75.5	29.6	71.4	28.2	67.8	27.1	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.5	8.5	-21.1	34.4	3.5	1.8	-23.7	35.6	-25.7	36.7	-27.7	37.7	-30.2	39.0		
(5)			-21.4	26.0	3.4	1.2	-23.9	26.9	-25.9	27.6	-27.8	28.3	-30.3	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-4.4	-3.2	1.8	8.6	14.9	20.3	22.4	21.5	17.6	10.8	4.3	-1.1		
	DBStd	10.59	5.79	5.68	6.15	5.09	4.77	3.73	3.13	2.97	4.14	4.69	4.89	5.05		
	HDD10.0	1751	445	371	265	86	10	0	0	0	1	47	181	345		
	HDD18.3	3603	703	603	513	296	130	22	4	8	63	238	421	602		
	CDD10.0	1577	0	1	12	42	161	309	386	356	228	72	10	1		
	CDD18.3	388	0	0	1	2	23	81	131	105	40	4	0	0		
	CDH23.3	3336	0	0	7	28	224	731	1191	791	327	37	0	0		
	CDH26.7	974	0	0	1	4	58	226	394	203	83	4	0	0		
(14) Wind	WSAvg	4.2	4.7	4.7	4.6	4.9	4.3	3.7	3.5	3.2	3.4	4.1	4.4	4.5		
(15) Precipitation	PrecAvg	943	55	44	64	93	88	95	88	90	96	84	81	65		
	PrecMax	1207	119	118	130	282	254	226	224	215	301	246	201	168		
	PrecMin	578	12	8	14	20	24	6	21	4	0	15	13	17		
	PrecStd	149	29	26	31	41	51	49	47	51	64	48	44	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.4	23.4	26.8	30.9	33.1	34.7	32.6	31.6	26.9	20.1	14.7		
		MCWB	11.5	11.1	16.4	17.0	20.5	23.3	24.7	23.7	21.3	20.0	14.4	12.6		
	2%	DB	8.6	10.0	18.5	23.2	28.4	31.2	32.1	30.6	29.0	23.9	16.6	11.0		
		MCWB	7.0	7.2	13.1	15.0	19.4	22.0	23.5	22.4	21.0	17.5	12.7	9.0		
	5%	DB	5.4	6.6	14.4	20.2	25.8	29.4	30.4	28.9	26.9	21.2	14.4	8.2		
		MCWB	3.8	4.1	10.5	13.0	18.0	21.2	22.5	21.5	19.8	16.5	11.0	6.2		
	10%	DB	3.2	4.1	11.0	17.5	23.5	27.5	28.8	27.4	25.0	18.8	11.9	5.4		
		MCWB	2.1	2.3	7.4	11.6	17.0	20.3	21.5	20.7	18.9	14.8	8.9	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.9	17.3	18.9	22.5	25.1	26.4	25.4	24.0	21.3	15.6	13.0		
		MCDB	12.6	13.4	22.9	24.9	28.5	30.7	31.4	30.6	28.8	25.7	18.1	14.2		
	2%	WB	7.3	7.4	14.2	16.2	20.7	23.5	25.0	24.0	22.5	19.2	13.5	9.6		
		MCDB	8.4	9.8	17.4	20.8	25.9	28.8	30.1	28.1	26.5	21.8	16.0	11.0		
	5%	WB	4.1	4.5	11.0	14.3	19.5	22.5	23.9	23.0	21.2	17.1	11.4	6.4		
		MCDB	5.1	6.3	13.8	18.9	24.1	27.6	28.8	26.7	25.0	20.4	13.9	7.8		
	10%	WB	2.0	2.4	7.6	12.1	17.9	21.4	22.8	22.1	19.9	15.2	9.1	3.8		
		MCDB	3.2	4.0	10.7	17.0	22.4	26.0	27.2	25.9	23.9	18.3	11.7	5.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.3	9.2	10.9	11.0	10.8	10.7	10.2	10.9	9.6	7.6	5.9		
		MCDBR	8.9	9.7	13.4	15.0	13.4	12.5	12.0	11.5	12.8	13.2	10.9	9.2		
	5% WB	MCWBR	7.9	7.9	9.0	8.5	6.9	5.7	5.4	5.3	6.1	7.9	8.0	7.8		
		MCWBR	8.3	9.2	12.2	13.0	11.6	10.5	10.5	9.5	10.4	11.5	9.9	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.346	0.342	0.378	0.411	0.431	0.421	0.422	0.384	0.348	0.316	0.295		
		taud	2.348	2.178	2.356	2.284	2.241	2.229	2.256	2.253	2.341	2.439	2.472	2.410		
	Ebn at Noon		837	835	899	888	864	845	849	835	843	834	813	807		
		Edh at Noon	81	114	109	127	136	138	133	129	110	88	73	71		
(44) All-Sky Solar Radiation	RadAvg		1.30	2.15	3.53	4.62	5.40	6.09	6.18	5.27	4.24	2.64	1.60	1.07		
	RadStd		0.12	0.28	0.38	0.44	0.54	0.39	0.29	0.30	0.40	0.30	0.22	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page